ModelIt!

Student Activity Pack

Can We 3D-Print a Human Organ?

Modeling Bioprinting, Tissue Engineering, and Transplant Viability

Grade 11L3 | NGSS HS-LS1-2, HS-ETS1-2 | Structure and Function

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you already know about 3D printing, and how do you think it might be adapted to print living tissue?**

2. **Why do you think there is a shortage of donor organs for transplantation, and what problems does this cause?**

3. **Draw a diagram showing how you think cells could be arranged in layers to build a piece of tissue, and label what each layer would need to survive.**

4. **What is the difference between printing a plastic object and printing living tissue — what additional challenges would living material create?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Print Resolution | Cell Viability Rate | Vascularization Depth | Scaffold Degradation Rate | Nutrient Diffusion Limit | Immune Compatibility | Functional Maturation

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Skin Patch Bioprinting

Moderate resolution, thin construct, no deep vascularization required

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Cardiac Patch with Vascularization

5mm thick cardiac construct requiring internal blood vessel channels

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Full Kidney Scaffold

Complex organ with millions of functional units and extensive vasculature

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Multi-Material Printing — The ability to deposit different cell types, scaffold materials, and growth factors from multiple print heads in a single print run — essential for organs that contain dozens of specialized cell types arranged in precise spatial patterns with different mechanical properties

Innervation — The growth of nerve fibers into bioprinted tissue to enable sensation and motor control — critical for functional integration of implanted constructs, particularly for tissues like skin that must sense touch and pain, or muscle that must respond to neural signals

Mechanical Conditioning — The application of controlled physical forces — stretching, compression, fluid flow — during tissue maturation to stimulate cells to produce stronger extracellular matrix and develop functional mechanical properties that match native tissue

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can we engineer living human organs from a patient's own cells — and could bioprinting end the transplant shortage that kills 17 people every day? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate why the 200-micrometer nutrient diffusion limit is the central constraint in organ bioprinting?**

2. **Why is matching scaffold degradation rate to tissue maturation rate so critical, and what happens when they are mismatched?**

3. **What does your model reveal about the qualitative difference between printing thin tissues and printing thick organs?**

4. **How would your model change if a method were developed to print functional capillary networks at the scale of individual cells?**

5. **What are the limitations of modeling bioprinting when individual patient cells behave differently and maturation rates vary unpredictably?**

8 STEM Challenge: Design a Bioprinted Tissue for Clinical Application

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YOUR ENGINEERING MISSION

Design a bioprinted tissue construct for a specific clinical application, addressing cell sourcing, scaffold design, vascularization strategy, maturation protocol, and implantation safety.

THE SCENARIO

A hospital's tissue engineering lab has asked your team to design a bioprinted tissue construct for one of three clinical needs: a skin graft for burn patients, a cartilage patch for knee repair, or a cardiac patch for heart attack recovery. Each application has different requirements for thickness, cell types, mechanical properties, vascularization, and functional integration. Your design must specify the bioink composition, print parameters, scaffold architecture, maturation protocol, and quality control tests to ensure the construct is safe and functional before implantation.

Engineering Design Thinking

1. WHAT BIOINK COMPOSITION — HYDROGEL TYPE, CELL DENSITY, GROWTH FACTORS — WOULD YOU SELECT FOR YOUR TISSUE TYPE AND WHY?

2. WHAT SCAFFOLD ARCHITECTURE WOULD PROVIDE THE MECHANICAL PROPERTIES YOUR TISSUE NEEDS WHILE ALLOWING CELL MIGRATION AND MATRIX REMODELING?

3. HOW LONG WOULD YOU EXPECT THE MATURATION PERIOD TO LAST AND WHAT ENVIRONMENTAL CONDITIONS WOULD YOU MAINTAIN IN THE BIOREACTOR?

4. WHAT ARE THE THREE MOST CRITICAL QUALITY CONTROL TESTS YOU WOULD PERFORM BEFORE APPROVING THE CONSTRUCT FOR IMPLANTATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Biomedical Engineers specializing in tissue engineering design bioprinting systems and tissue constructs at companies like Organovo, CELLINK, and academic medical centers, earning $80,000-$150,000/year. Regenerative Medicine Scientists develop new approaches to growing replacement tissues and organs, working at institutions like the Wake Forest Institute for Regenerative Medicine, earning $90,000-$170,000/year. Clinical Tissue Engineers oversee the production and quality control of bioprinted constructs for patient use in hospital settings, earning $85,000-$140,000/year.